

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE						
Supplementary Winter Examination – 2024						
Course: B.Tech		Branch : Chemical Engineering		Semester : Sixth		
Subject Code & Name: Heat Transfer Equipment Design (BTCHO605B)						
Max Marks: 60		Date:30/12/2024		Duration: 3 Hr.		
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.						
					(Level/CO)	Marks
Q.1	Objective type questions. (Compulsory Question)					12
1	Which of the following is NOT a type of heat exchanger?				CO1	1
	a. Reaction vessel	b. Shell and tube heat exchanger	c. Plate heat exchanger	d. Air-cooled heat exchanger		
2	In heat exchangers, the primary purpose of baffles is to:				CO1	1
	a. Direct flow to increase turbulence.	b. Reduce pressure drop	c. Prevent fouling	d. Increase the heat transfer surface area		
3	Which of the following fluids would typically require a higher heat transfer area due to its low thermal conductivity?				CO2	1
	a. Oil	b. Ammonia	c. Air	d. Water		
4	Which parameter determines the effectiveness of a heat exchanger?				CO3	1
	a. Overall heat transfer coefficient.	b. Fouling factor	c. Temperature gradient	d. Heat exchanger geometry		
5	The log-mean temperature difference (LMTD) is used in heat exchanger design to account for:				CO4	1
	a. Temperature differences across the	b. Flow type (parallel or counterflow)	c. Heat transfer coefficients	d. Fouling effects		

	exchanger					
6	Which factor is most likely to cause thermal stress in heat exchanger components?				CO5	1
	a. Large temperature differences	b. High flow rates	c. Low fouling factor	d. Material homogeneity		
7	Fouling in heat exchangers results in:				CO2	1
	a. Reduced heat transfer efficiency	b. Increased overall heat transfer coefficient	c. Decreased pressure drop	d. Improved fluid mixing		
8	In evaporators, the boiling point elevation is an important consideration when:				CO1	1
	a. The fluid contains dissolved solids	b. The fluid has a high vapor pressure	c. The pressure of the system is low	d. The fluid is a pure substance		
9	Which type of heat exchanger is most suitable for handling two-phase heat transfer, such as condensation?				CO4	1
	a. Shell and tube heat exchanger _____	b. Air-cooled heat exchanger _____	c. Double-pipe heat exchanger _____	d. Plate heat exchanger _____		
10	What happens to the heat transfer rate if the flow velocity of a fluid in a heat exchanger increases?				CO3	1
	a. Increases due to reduced boundary layer thickness	b. Decreases due to higher turbulence	c. Remains constant	d. Increases due to higher thermal conductivity		
11	In a shell and tube heat exchanger, the fluid in the tubes is referred to as the:				CO2	1
	a. Tube-side fluid	b. Shell-side fluid	c. Working fluid	d. Process fluid		
12	The purpose of baffles in a shell and tube heat exchanger is to:				CO2	1

	a. Both a and b	b. Reduce fouling	c. Direct the shell-side fluid flow	d. Support the tubes		
Q.2	Solve the following.					12
A)	What is the purpose of a heat exchanger in process industries?				CO1	6
B)	Explain the differences between counterflow, parallel flow, and crossflow heat exchangers.				CO2	6
Q.3	Solve the following.					12
A)	What are the primary factors that influence the selection of heat exchanger type for a specific application?				CO2	6
B)	Define the term "overall heat transfer coefficient" and explain its significance in heat exchanger design.				CO3	6
Q.4	Solve Any Two of the following.					12
A)	Hot oil at a rate of 12 kg/s [$C_p = 2083 \text{ J/(kg}\cdot\text{K)}$] flows through double pipe heat exchanger. It enters at 633 K (360°C) and leaves at 573 K (300°C). The cold fluid enters at 303 K (30°C) and leaves at 400 K (127°C). If the overall heat transfer coefficient is $500 \text{ W/(m}^2\cdot\text{K)}$, calculate the heat transfer area for (i) parallel flow and (ii) countercurrent flow.				CO2	6
B)	Explain in brief on double pipe heat exchanger.				CO3	6
C)	Explain briefly various types of condenser.				CO5	6
Q.5	Solve Any Two of the following.					12
A)	An evaporator operating at atmospheric pressure (101.325 kPa) is fed at the rate of 10000 kg/h of weak liquor containing 4% caustic soda. Thick liquor leaving the evaporator contains 25% caustic soda. Find the Capacity of the evaporator.				CO6	6
B)	Define the following terms: (i) Evaporation (ii) Boiling Point Elevation (iii) Capacity of Evaporator And (iv) Economy of an Evaporator.				CO5	6
C)	Explain in brief different types of agitators.				CO6	6

Q. 6	Solve Any Two of the following.		12
A)	Draw the neat sketch of kettle Reboiler and explain in brief its construction.		6
B)	Explain the role of pinch analysis in process heat integration.		6
C)	What are the design challenges when handling corrosive fluids in heat exchangers?		6
	*** End ***		